EUROPEAN UNION STEM LABORATORY INITIATIVE

SPHERE: MISSION CONTROL

TEACHER MANUAL & SOLUTIONS KEY

Inquiry-Based Science Education (IBSE) Framework: This instructor manual provides curriculum alignment, a 60-minute lesson plan, equipment preparation protocols, theoretical physics decay reference tables, and complete official solutions for the Student Worksheet & Question Sheet (Q1 through Q8).

1. CURRICULUM CONTEXT & SYLLABUS ALIGNMENT

The **SPHERE: Mission Control Digital Twin Laboratory** integrates hands-on experimental physics with computational modeling. Designed for secondary and upper-secondary physics (NGSS, IB Physics, A-Level, and National STEM Curricula), this module targets the following core learning outcomes:

- **Heat Transfer Mechanisms:** Conduction, Convection, and Thermal Radiation.
- **Exponential Mathematical Decay:** Differential and integrated Newton's Law of Cooling $T(t) = T_{env} + (T_0 - T_{env})e^{-kt}$.
- **Digital Twin Engineering Methodology:** Validating physical hardware sensors against computer simulation models.
- **Data Analysis & Graphing Literacy:** Manual data logging, graph construction, and percentage error evaluation.

2. LESSON PLAN & TIMELINE (60 MINUTES)

TIMEFRAME	LESSON PHASE	TEACHER & STUDENT ACTIVITIES
00–10 min	1. Mission Briefing & Quiz	Teacher introduces orbital thermal challenges. Students complete the Pre-Flight Quiz in Mission Control to unlock telemetry features.
10–20 min	2. Hardware Capsule Assembly	Student crews wrap 100ml PET jars with assigned insulation (Bare, Cotton, Mylar, MLI) and thread sensors through PG7 waterproof glands.
20–35 min	3. Live 15m Telemetry Run	Capsules submerged in 0°C ice bath. Students record physical core temperatures every 60s into Mission Control & Task Sheets.
35–48 min	4. Graphing & Correlation	Crews plot paper graph curves, compute absolute errors $\lvert\Delta T\rvert$, and calculate their crew Thermal Correlation Score.
48–60 min	5. Debrief & Analysis	Instructor leads reflection on material decay rates (k), sources of experimental error, and spacesuit engineering trade-offs.

3. EQUIPMENT PREPARATION & LAB PROTOCOLS

To ensure smooth lab execution, prepare materials prior to student arrival:

- **Water Temperature Control:** Maintain hot water supply at 80°C (use a thermostatically controlled kettle).
- **Ice Bath Preparation:** Prepare slurry buckets containing 50% crushed ice and 50% cold water to guarantee stable T_{env} = 0°C boundary conditions.
- **PG7 Seal Verification:** Demonstrate tightening the PG7 gland dome nut to prevent liquid leakage into student capsules.

4. THEORETICAL PHYSICS REFERENCE & DECAY CONSTANTS

Expected end temperatures (t = 15 min) based on T_0 = 80°C and T_{env} = 0°C:

INSULATION SYSTEM	DECAY RATE (k)	EXPECTED TEMP (15m)	DOMINANT THERMAL DISSIPATION PATHWAY
A. Bare Control	k ≈ 0.150 / 0.082	8.4°C – 23.3°C	Rapid uninsulated liquid conduction & convective fluid currents.
B. Cotton Layer	k ≈ 0.080	24.1°C	Traps stationary air cells to reduce conductive losses.
C. Mylar Layer	k ≈ 0.040	43.9°C	Reflects infrared radiant electromagnetic heat energy.
D. Full MLI System	k ≈ 0.015	63.9°C	Multi-layer barrier blocking conduction, convection, and radiation.

SECTION 5: QUESTION SHEET OFFICIAL SOLUTIONS KEY (Q1–Q8)

INSTRUCTOR SOLUTIONS

Q1 Solution: Thermal Radiation vs Conduction in Liquid

Exemplar Response: Mylar is an ultra-thin metallic film optimized to reflect infrared radiation. Radiation is the dominant form of heat transfer in space vacuums. However, in an ice-water bath, conduction via direct liquid contact is extremely fast. Because Mylar is very thin and highly conductive to touch, it easily transfers heat directly to surrounding water if used alone. To be effective in a liquid bath, it requires an inner layer (like cotton or bubble wrap) to block conduction.

Q2 Solution: Physical Meaning of Cooling Constant (k)

Exemplar Response: A **smaller** k value represents a **better** thermal insulator. In Newton's exponential equation $T(t) = T_{env} + (T_0 - T_{env})e^{-kt}$, the decay exponent is proportional to $-k \cdot t$. As k approaches zero, the exponential factor e^{-kt} stays near 1.0, keeping core temperatures high and delaying heat loss over time.

Q3 Solution: Multi-Layer Insulation (MLI) Synergy

Exemplar Response: MLI combines three complementary barriers: cotton traps microscopic stationary air pockets to block conduction; bubble wrap creates sealed air chambers that suppress fluid convection currents; and metallic Mylar reflects thermal radiation back into the core. By simultaneously targeting all three modes of heat transfer, the combination achieves a dramatically lower decay rate (k = 0.015) than any single material alone.

Q4 Solution: Physical vs Digital Twin Discrepancy Sources

Exemplar Response (Accept any two):

1. Sensor placement: probe tip touching container walls instead of hanging freely in water core.
2. Incomplete gland tightening leading to cold water seepage into capsule.
3. Thermal stratification: unmixed water inside jar creating localized temperature gradients.
4. Fluctuations in ice bath temperature above 0°C during prolonged testing.

Q5 Solution: Environmental Boundary Calibration

Exemplar Response: Students should measure the actual ambient ice bath temperature using their DS18B20 thermometer and enter that exact value (e.g., T_{env} = 4.5°C) into the T_{env} parameter field in Mission Control. Updating the environmental boundary parameter ensures the mathematical digital twin accurately models physical classroom conditions.

Q6 Solution: Water Volume & Heat Capacity Scaling

Exemplar Response: Doubling the water volume from 100ml to 200ml doubles the total thermal mass m. Because thermal energy stored is $Q = m \cdot c \cdot \Delta T$, the core holds twice as much thermal energy. For the same surface area, the temperature drop per minute will be slower, resulting in a **smaller effective decay rate constant *k***.

Q7 Solution: Spacesuit Design Trade-Offs in EVA

Exemplar Response: In real space missions, adding excessive insulation layers increases suit bulk, mass, and stiffness. Astronauts require high joint flexibility and mobility during Extravehicular Activities (EVAs). Excessively thick suits also restrict movement and increase metabolic heat accumulation inside the suit, requiring heavy active cooling loops (LCVG garment).

Q8 Solution: Correlation Score Optimization

Exemplar Response: Crews can optimize scores by: (1) Ensuring the PG7 gland dome nut is fully hand-tightened to eliminate liquid leaks; (2) Centering the probe tip in the liquid core; (3) Gently swirling the capsule before reading to eliminate thermal stratification; (4) Standardizing timer readings to the exact second.

6. ASSESSMENT RUBRIC & GRADING MATRIX

CRITERIA	EXEMPLARY (A / 90-100%)	PROFICIENT (B / 75-89%)	NEEDS RE-CALIBRATION (<75%)
Data Logging	Complete 15-min table with accurate sensor logging & detailed notes.	Missing 1-2 time intervals; basic notes logged.	Multiple blank intervals; incomplete data tables.
Graph Accuracy	Precise points plotted; clean solid & dashed lines; fully labeled axes.	Points plotted with minor curve scaling inaccuracies.	Misaligned axes; points not connected; missing legend.
Physics Analysis	Thorough answers addressing conduction, convection, radiation, MLI, and volume scaling (Q1–Q8).	Correct identification of heat transfer modes with minor details missing.	Incorrect concepts; confuses radiation with conduction.